

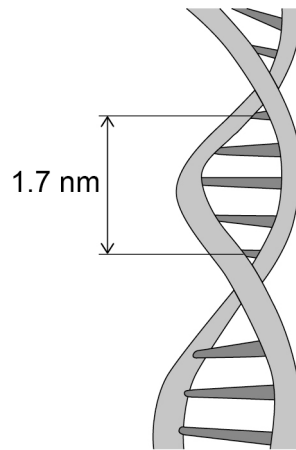
Answer **all** questions in the spaces provided.

Do not write
outside the
box

0 1

Figure 1 shows part of a DNA molecule.

Figure 1



0 1 . 1

Name the type of bond between:

[2 marks]

complementary base pairs _____

adjacent nucleotides in a DNA strand _____

0 1 . 2

The length of a gene is described as the number of nucleotide base pairs it contains.

Use information in **Figure 1** to calculate the length of a gene containing 4.38×10^3 base pairs.

[2 marks]

Answer _____ nm



0	1	.	3
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Describe **two** differences between the structure of a tRNA molecule and the structure of an mRNA molecule.

[2 marks]

1 _____

2 _____

0	1	.	4
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In a eukaryotic cell, the structure of the mRNA used in translation is different from the structure of the pre-mRNA produced by transcription.

Describe **and** explain a difference in the structure of these mRNA molecules.

[2 marks]

8

Turn over for the next question

Turn over ►

